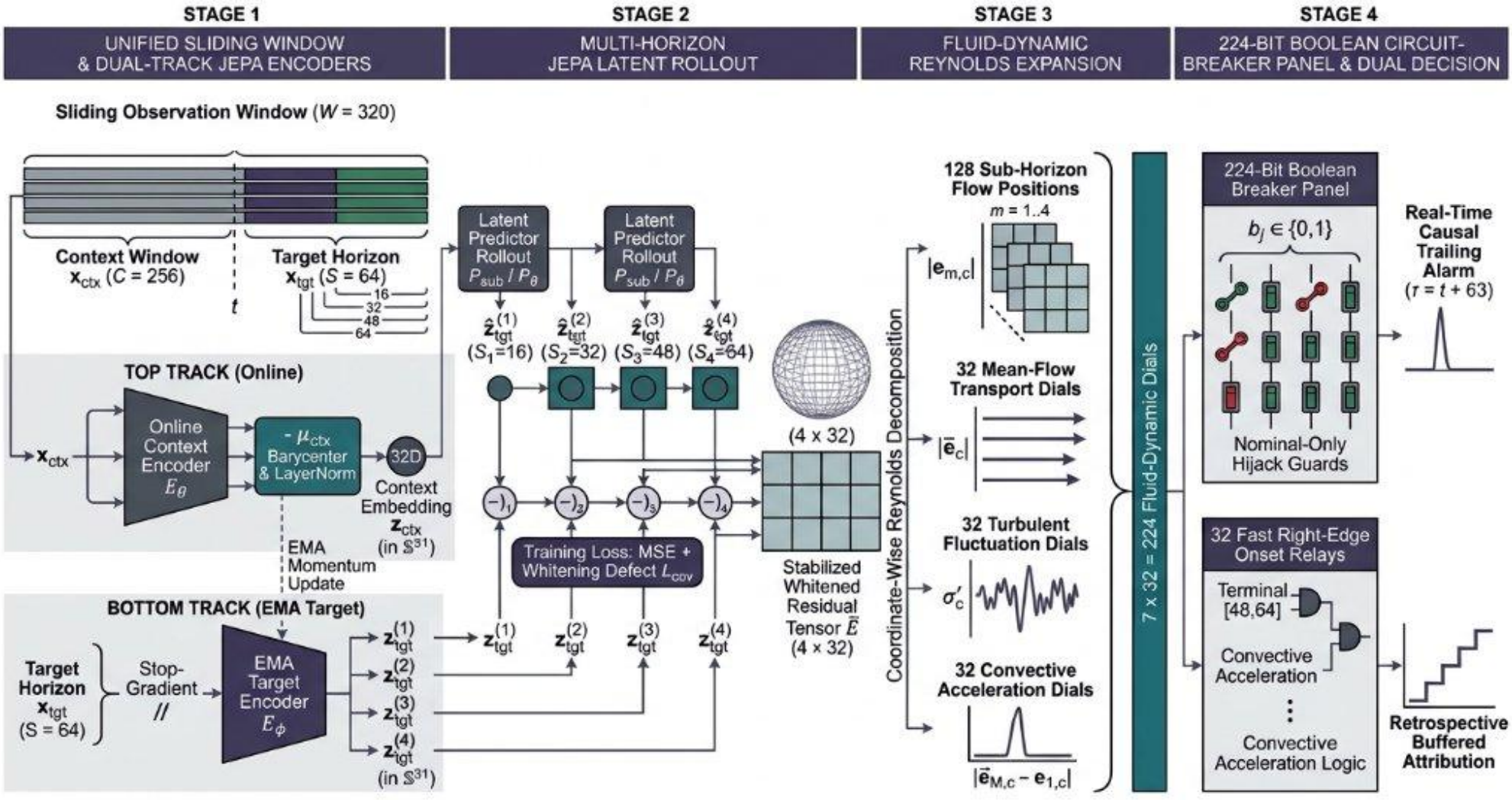

STAGE 1
UNIFIED SLIDING WINDOW & DUAL-TRACK JEPA ENCODERS
STAGE 2
MULTI-HORIZON JEPA LATENT ROLLOUT
STAGE 3
FLUID-DYNAMIC REYNOLDS EXPANSION
STAGE 4
224-BIT BOOLEAN CIRCUIT-BREAKER PANEL & DUAL DECISION
Sliding Observation Window (W = 320)
Context Window x_ctx (C = 256)
Target Horizon x_tgt (S = 64)
16
32
48
64
t
TOP TRACK (Online)
x_ctx
Online Context Encoder E_θ
- μ_ctx Barycenter & LayerNorm
32D
Context Embedding z_ctx (in S^31)
EMA Momentum Update
BOTTOM TRACK (EMA Target)
Target Horizon x_tgt (S = 64)
Stop-Gradient //
EMA Target Encoder E_ϕ
z_tgt^(1)
z_tgt^(2)
z_tgt^(3)
z_tgt^(4)
(in S^31)
Latent Predictor Rollout P_sub / P_θ
Latent Predictor Rollout P_sub / P_θ
ẑ_tgt^(1) (S_1=16)
ẑ_tgt^(2) (S_2=32)
ẑ_tgt^(3) (S_3=48)
ẑ_tgt^(4) (S_4=64)
Training Loss: MSE + Whitening Defect L_cov
(4 x 32)
Stabilized Whitened Residual Tensor Ē (4 x 32)
Coordinate-Wise Reynolds Decomposition
128 Sub-Horizon Flow Positions
m = 1..4
|e_m,c|
32 Mean-Flow Transport Dials
|ē_c|
32 Turbulent Fluctuation Dials
σ'_c
32 Convective Acceleration Dials
|ē_M,c − e_1,c|
7 x 32 = 224 Fluid-Dynamic Dials
224-Bit Boolean Breaker Panel
b_j ∈ {0,1}
Nominal-Only Hijack Guards
Real-Time Causal Trailing Alarm (τ = t + 63)
32 Fast Right-Edge Onset Relays
Terminal [48,64]
Convective Acceleration
Convective Acceleration Logic
Retrospective Buffered Attribution